Every Press Counts

Weight from stroke counts for Europress SMART

Europress challenge, Since AI Hackathon 2026, Turku. Team THEAGLEYE. 8 November 2026.

Part 1: Getting the weight

Part 2: What the weight is worth, how it runs in SMART, and what Europress can build on it

What this work rests on, and what it does not show

All data is synthetic, and all euro figures use the dataset's hypothetical prices. Code, results and the demo page are in the repository.

Part 1: Getting the weight

The idea

A Europress compactor already knows how many times its ram has pressed. Every stroke pushes one charge of waste from the hopper into the container, so the number of strokes since the last lift says how much waste has gone in. Multiplying that count by the kilos one stroke carries at that site gives the weight in the container:

```
kg = strokes since the last lift x kg per stroke for this site x 1.1
```

The factor 1.1 makes the estimate read about 10% high on purpose. A high reading only moves a lift slightly earlier, while a low reading lets the container overflow, so the error is pushed to the side that costs less. Europress's own simulator confirms this: a gauge that reads 6% high is worth €258 per machine-year, one that reads 6% low only €208 (`results/sim_value.csv`).

A stroke is a press, not a weight. How much one press moves depends on the people using the machine: one site fills the hopper before pressing, another presses after every single box. For this reason each site is taught its own kilos per stroke once, from 3 weighed lifts. After that the estimate runs in the SMART cloud from data the controller already sends, with no permanent scale and no daily operator data.

The concept assumes that real SMART controllers log the stroke count, as the `compaction_cycles` field in the dataset does. The controller starts every stroke, so a machine generation that does not report the count needs a firmware change, not new hardware. This has not been confirmed with Europress.

Teaching a site

Each site starts from the average kilos per stroke of its waste stream, learned from a few reference machines. In the official dataset these averages are 31.9 kg for biowaste, 27.7 kg for mixed waste, 21.7 kg for paper, 20.6 kg for cardboard, 9.9 kg for plastic film and 8.9 kg for mixed plastic (`results/kg_per_stroke.csv`).

The site's own value comes from its first 3 weighed lifts: the median of weighed kilos divided by strokes. There are two ways to get those weights. The cheaper way is to ask the hauler once, at setup, for the weighbridge tickets of 3 lifts. That needs no hardware and no visit, and in the dataset 13,442 of 18,297 completed lifts already have a ticket. Where the hauler does not share, a Europress technician brings a mobile calibration kit: load pads under the container, calibrated with test weights in the factory, left in place for 3 fills and then moved to the next site. The kit has not been built or tested.

Habits change when staff change, with the seasons or as the machine wears. A drift alarm watches for this by comparing the stroke estimate with an estimate from the fill reading. When the two disagree by more than 2 times for 3 lifts in a row, SMART asks for one new weighing and uses the fill reading until it arrives. In the stress test below this asked for 0.07 to 0.09 re-weighs per machine-year at sites with a stable habit.

How the estimate checks itself

The stroke count measures mass and the fill reading measures volume, so each can check the other. In a common compactor design the ram runs a fixed stroke and the back pressure at the end of the stroke rises as the container fills (US4643087). The ram travel shortens only near full, so the fill estimate, not the ram travel, is the volume signal. Four checks follow from this.

- **Telemetry coverage.** Missing hourly messages mean missing strokes, which makes the estimate read low. When fewer than 95% of the expected messages have arrived since the last lift, the estimate is marked low confidence and SMART falls back to the fill reading at 90%, the same fallback Europress's simulator uses.
- **Capacity.** A container cannot hold much more than its capacity in kilos, which the dataset's material table gives as volume x compacted density x (1 + compressibility / 2). An estimate above 1.3 times capacity is an over-read, usually strokes from more than one fill, and is marked low confidence. It never blocks a lift call, because the estimate passes the 80% call long before it reaches 1.3 times capacity.
- **Stuck fill sensor.** A fill reading of 85% or more while the strokes say less than 40% of capacity means the fill sensor is stuck. SMART ignores the fill reading and opens a service ticket.
- **Drift.** A lasting disagreement between strokes and fill means the site's habit changed, so SMART asks for a new weighing as described above.

How accurate it is

`analysis/strokes_eval.py` tests the stroke count on 5,383 lifts of 368 machines with a weighbridge ticket, always using kilos per stroke learned on other machines. The stroke count has a median error of 7.5%, against 13.0% for an estimate from fill x density. More important, it rarely reads low: 2.5% of lifts are underestimated by more than 20%, against 13.7% for fill x density, and the 1.1 setting brings this down to 0.8%.

The error falls as more strokes go into one lift, because the random differences between single charges average out. Lifts with 11 to 50 strokes have a median error of 10.8%, while lifts with 101 to 200 strokes have 6.0%. Biowaste, with a median of 41 strokes per lift, and paper, with 56, sit in the noisiest range. Cardboard, with 115, sits in the best one.



The worst errors come from two sources. Lifts with telemetry gaps, 3.4% of all lifts, are underestimated by more than 20% in 19.1% of cases, against 1.9% with full telemetry, which is why the coverage check exists. The capacity check flags 393 lifts (7.3%) as over-reads, and after it the median error is 6.9% and the 90th percentile drops from 61.0% to 24.7%.

What accuracy each use needs

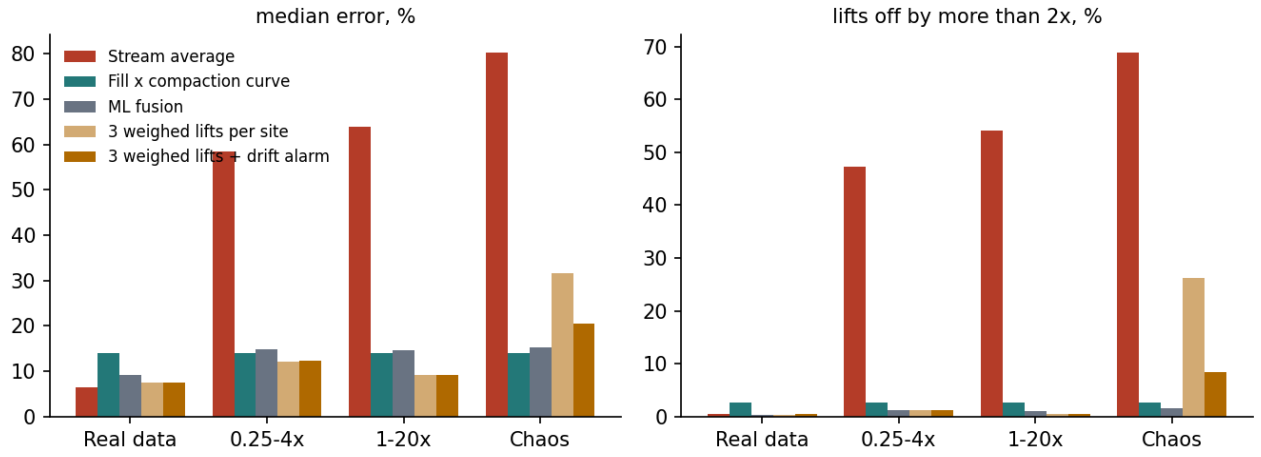
The brief's simulation target is about ±100 kg on a 3 t cardboard load. The stroke count does not meet it: only 34.0% of 562 loads between 2.5 and 3.5 t are within ±100 kg, against 77.9% for the dataset's simulated built-in scale (`results/accuracy_use_cases.csv`). Different uses need different accuracy, so the concept uses the stroke count only where it is good enough.

Deciding when to lift needs a ballpark. In Europress's simulator the value of lifting by kilos stays flat up to about 9% median error, so the stroke count at 6.9% to 7.5% is enough. A cost per tonne report for the customer is also fine, with a median error of 7.0% in €/t. Checking an operator's invoice is different: Europress's own example script catches 94.7% of a 10% tonnage overbilling with a 1.5% scale, but only 15.6% with a 10% error (`results/europress_accuracy_example.txt`). So invoice checks stay with reference gauges or weighbridge tickets.

Stress test: what if every site presses differently?

The dataset gives every site the same pressing habit, which flatters any method that uses one average per stream. `analysis/habits.py` breaks the real stroke counts on purpose and then tests ways to recover the weight. Each site gets a random habit, applied hour by hour, in three scenarios: each site presses 0.25 to 4 times as often per kilo, each site presses 1 to 20 times as often (tiny charges, one box per press), and a chaos case where habits also change up to 3 times a year and empty and idle presses are added. In the chaos case the strokes per machine end up a median 4.3 times the original, and up to 114 times.

In each of 5 rotations, 20% of the machines act as reference machines that teach and the other 80% are scored against their weighbridge tickets, which gives 21,624 scored lifts per scenario. Every estimate uses only the machine's past cycles.



One average per stream does not survive. With different habits, 47% to 69% of lifts are off by more than 2 times. The fill reading, through a compaction curve learned on the reference machines, is not affected by habits at all, but it stays at 14.0% median error. A Kalman filter that learns each site's kilos per stroke from the fill reading ended at 13.6% to 22.2%, no better than the fill reading itself, because the fill sensor's own per-machine bias limits it. Gradient boosting that fuses strokes, fill and the Kalman estimate is the best method that needs no weighing at the site, at 14.6% to 15.2% with only 1% to 2% of lifts off by 2 times.

Teaching each site with 3 weighed lifts brings the error back to 9.1% when sites press 1 to 20 times differently, with 0.4% of lifts off by 2 times. Weighing 1, 3 or 5 lifts gives 10.7%, 9.1% and 8.8%, so the gain flattens after 3. When habits keep changing, the taught value goes stale and the error rises to 31.7%. The drift alarm brings it down to 20.6%. An earlier, stricter alarm setting (1.5 times for 2 lifts) reached 16.5%, but it asked for 0.63 to 1.87 re-weighs per machine-year, and the costing below shows that this would eat most of the value.

Two more methods were tried and not adopted. Conformal prediction gave each fused estimate a 90% range, but the ranges covered only 86% to 87% of true weights and spanned about 2 times from low to high. Inverse-variance fusion of the Kalman estimate and the fill curve landed at 12.6% to 14.2%, close to the fill reading alone.

The habits are injected by this work, not observed, and only a pilot can show how far real sites differ.

Retrofit and cost

Three retrofit paths cover the installed base. SMART-connected machines whose controller already reports strokes need only the cloud software. Connected machines that do not report strokes need a firmware release per controller generation, which the controller can take remotely if it supports remote updates. Machines that are not connected, or of another brand, need a telemetry retrofit unit plus a stroke counter, for example a current sensor on the pump motor supply where each stroke shows as one load peak. Counting

compactor strokes from motor energy is described in KAPASITY Oy's application WO2021176137A1, where the count serves fullness, not weight, and motor current per cycle was used for fullness already in US5558013A. The third path is not tested in this work.

The cost of teaching one site with the kit, without operator data, adds up from five figures. The kit is priced from one listed roll-off container scale with four load cells, €4,800 to €5,620 excluding VAT (the listing shows both prices). Three fills take a median of 21.0 days in the replay, so one kit could teach up to 17 sites a year, and 12 is used as a realistic lower figure. Over an assumed 5-year life that is €55 to €94 per site. Two technician visits of an assumed 1.5 hours each cost €117 at the average Finnish private-sector labor cost of €39.0 per hour in 2024 (Statistics Finland). Driving the dataset's median site round trip of 28 km twice costs €30.8 at the 2026 tax-free allowance of €0.55 per km. Teaching a site costs about €203 to €242 once.

A re-weigh after a drift alarm costs about €169: the same two visits plus 7 days of kit use. With the drift alarm at 2 times for 3 lifts, stable sites need 0.07 to 0.09 re-weighs per machine-year, about €12 to €15 a year. A chaotic site needs 0.57, about €97 a year. Against €225 per machine-year of simulated value, a taught site pays back in about 1 to 1.15 years. A permanent scale costing €4,800 to €5,620 per machine would need 20 to 23.5 years to pay back from the €239 a real gauge adds, before installation.

The software is the main fixed cost. Its effort is not known: one person-month costs about €6,240 at the same average labor cost, and software developers likely cost more. If a production feature took 6 to 12 person-months, that would be €37,000 to €75,000, but this is a guess, not an estimate from data. Kit engineering for DUO rails and COMBI feet, transport of kits and cloud compute are not priced.

Reference gauges

A few machines per stream and region carry a permanent gauge as reference instruments. On a COMBI, where the truck lifts the whole unit, load pins in the unit's feet or roller axles travel with it and read zero when lifted, which gives a free zero check at every pickup. On a DUO, where the container slides onto rails at the compactor, load cells sit under the rails and need stops against the pulling force of the hookloader. Load-cell weighing of compactor containers is established prior art (BACE, US10564029B2; Canalytics, US12379240B1), so the gauge is not the new part. It is the instrument that teaches the stream averages and checks tonnage where a customer needs it.

Sources of error

Most errors fall into two groups. Some make the estimate read low, which is the harmful direction: lost telemetry, a pallet or other heavy item in one stroke, and rain or wet waste that adds mass without strokes. The coverage check and the 98% fill stop protect against overflow from these, while heavy charges and rain are not tested. Others make it read high, which costs an early lift: extra, empty or idle presses, a partial lift that does not reset the count, and a stuck fill sensor that orders lifts. Per-site teaching, the capacity check and

the watchdog handle these. A wrongly configured waste stream gives a wrong start value, which the site's own weighings replace.

Validation plan

1. **Desk check, one day.** Repeat `analysis/strokes_eval.py` on Europress's real cycle counts and any weighbridge tickets customers already share. The pass mark is a median error at or below 9% per stream.
2. **Pilot, 8 to 12 weeks.** 20 to 30 machines in plastic film, cardboard and mixed waste, where the value is highest. Each site is taught with 3 weighed lifts, then every lift is weighed by one cooperating operator or the kit. The key real-world numbers are how far kilos per stroke differ between sites and how stable they stay at one site.
3. **Metrics.** Median error, 90th percentile error, signed bias, the share of lifts more than 20% low and more than 20% high, and re-weighs per machine-year. Machines being scored never teach.
4. **Proposed targets.** Median error at or below 9%, and at most 3% of lifts more than 20% low after the 1.1 setting. These are targets, not results.
5. **Decision test.** Run the kilo trigger against the fill trigger on matched pilot sites and compare lifts per tonne and overflow events.

Part 2: What the weight is worth, how it runs in SMART, and what Europress can build on it

All figures come from Europress's synthetic dataset and public generator, with the dataset's hypothetical prices. Each figure names the script and the result file it comes from.

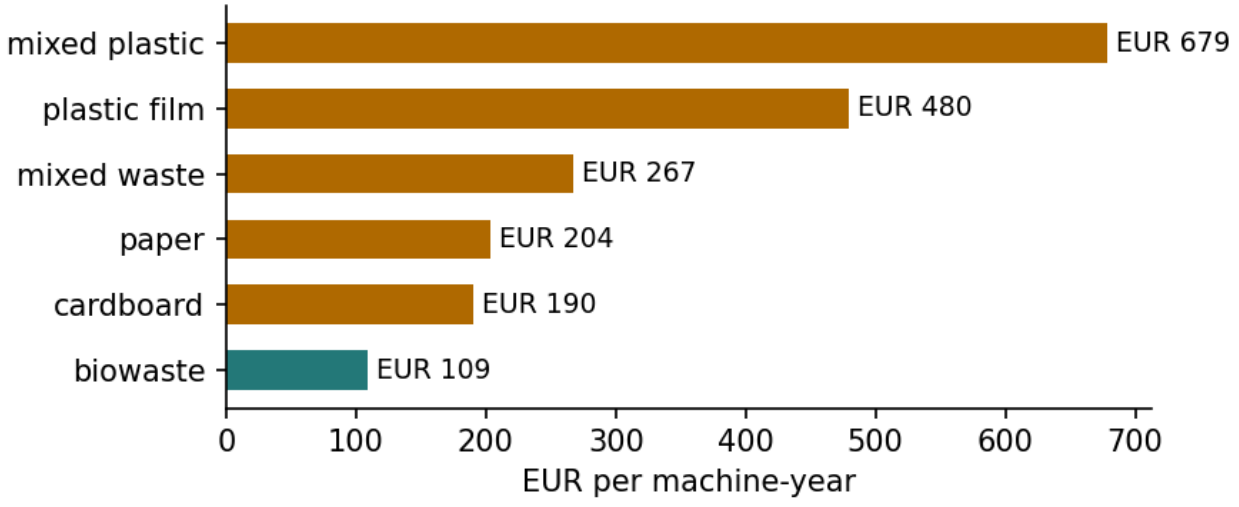
Where the money is lost today

Customers pay a fixed fee for every lift, so a lift that comes early charges a full fee for a partly empty container. The official labels count 1,854 such unnecessary lifts between 1 October 2025 and 30 June 2026, worth €274,760 (`results/premature_lifts.csv`). Most of them, 1,216 lifts and €171,037, are plain over-servicing. Another 295 lifts and €53,328 come from fill sensors stuck at a high reading, and 343 lifts and €50,395 from precautionary orders.

The fill reading causes much of this. In Europress's own model every machine has its own gain and offset error, and plastics spring back after pressing, so their fill reads 4 to 5 points high. A lift ordered at a fill threshold then comes before the container is full in kilos.

What lifting by kilos adds

Europress's decision simulator offers a kilo trigger that lifts at 80% of the container's capacity in kilos. On the 3,573 official decision cases it adds 23.4% on top of the best decision available without weight, which is €262 per machine-year across all streams (`results/value_by_stream.csv`). The gain follows the fill error. Mixed plastic gains €679 and plastic film €480 per machine-year, because their fill reads high, while biowaste gains only €109.

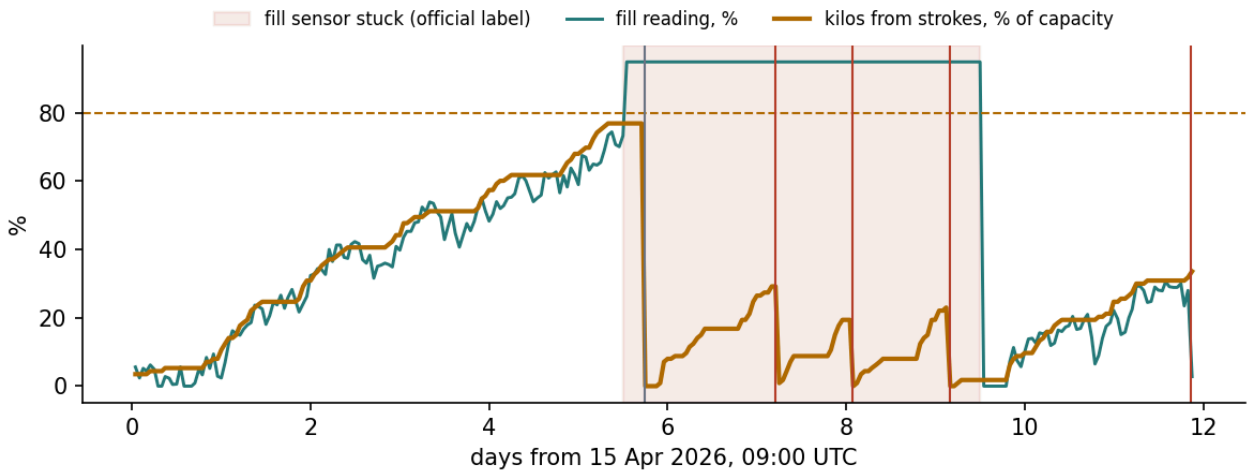


For one customer the gain adds up over its machines (`results/value_by_customer.csv`). The 52 customers have a median of 3 machines with decision cases, and the median customer

gains €672.5 a year. The top tenth gain €5,533 or more, and the largest customer, with 54 machines, gains €11,431.

The sensor that froze: one machine, five days

CMP-00168 is a 12 m³ DUO cardboard compactor belonging to customer ORG-0023, whose 31 machines would gain €7,112 a year. On 20 April 2026 at 21:00 UTC its fill sensor stuck at 94.9% and stayed there for four days, which the official label calls a false-high fill reading caused by bridging or obstruction (`results/replay_trace.json`).



SMART kept trusting the stuck reading and ordered four lifts within five days. The strokes told a different story: they put the container at 748 kg, 499 kg, 589 kg and then 816 kg, between 19% and 32% of its 2,565 kg capacity, and the dataset's own gauge agreed with 620 kg, 495 kg, 475 kg and 695 kg. All four lifts are labeled unnecessary, with €871.19 at stake. SMART Weight would have held all four. Its watchdog flagged the fill sensor on 21 April at 09:00, 29 hours before the first unnecessary lift, and kept flagging it after each lift until the sensor recovered.

One lift traced from strokes to euros

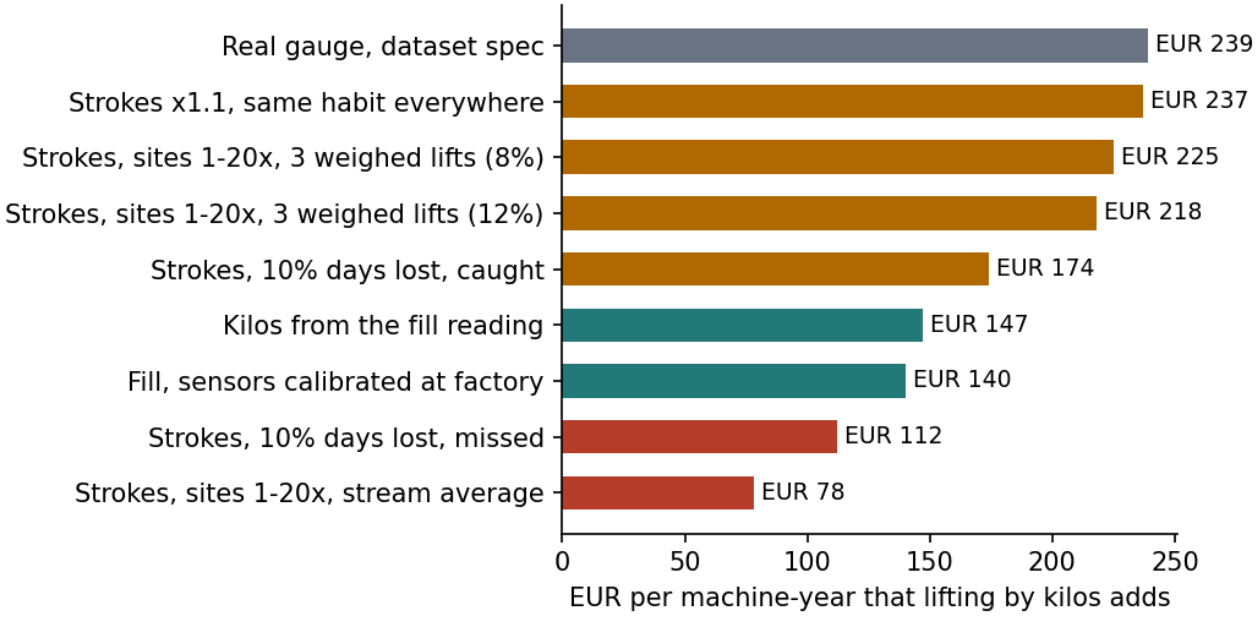
Lift COL-00168-0034 on 23 April shows how each number follows from the one before it.

1. The machine reported 22 strokes in the 20 hours since its previous lift (`telemetry_dedup.compaction_cycles`, `results/replay_cycles.csv`).
2. Cardboard carries 20.6 kg per stroke (`results/kg_per_stroke.csv`), so with the 1.1 setting the estimate is 22 x 20.6 x 1.1 = 498.5 kg.
3. The 12 m³ container holds 12 x 190 x (1 + 0.25 / 2) = 2,565 kg of cardboard (`configs/assumptions.yaml`), so it was 19.4% full in kilos, far below the 80% call.
4. Telemetry was complete and the estimate was below 1.3 times capacity, so confidence was normal.
5. The fill reading said 94.9% while the kilos said under 40%, so the watchdog had marked the fill sensor as stuck.
6. The dataset's gauge read 495 kg before the lift, 0.7% below the estimate.

7. The official label marks the lift unnecessary because of the false-high fill reading, with €225.44 at stake (`anomaly_labels`).

How good the weight has to be

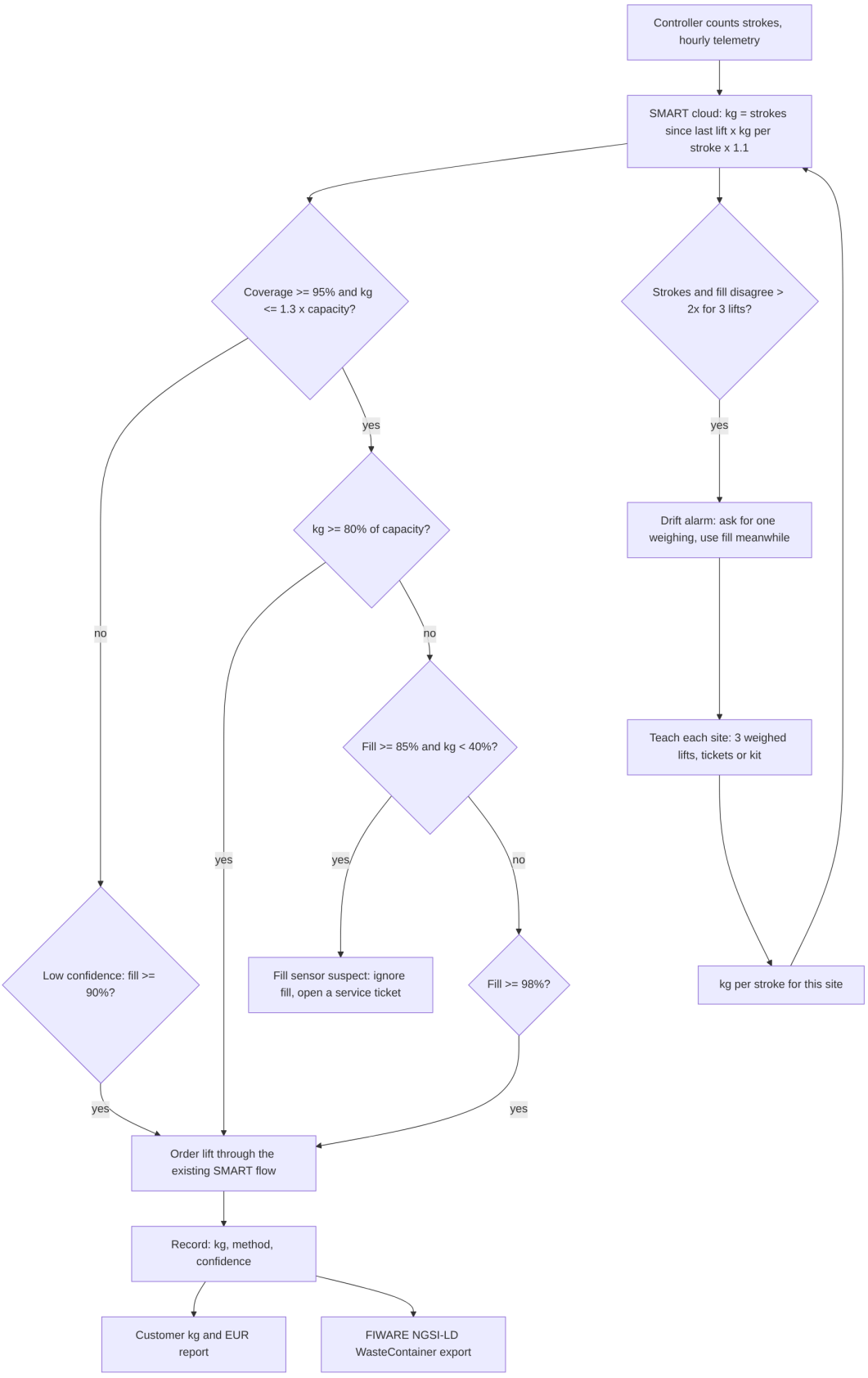
`sim/experiments.py` rebuilds Europress's world 25 times with its public generator (seed 31337, 250 machines, one year, 1,851 decisions) and changes one thing per run. The best value without weight is €142,557 in every run, so only the tested change differs (`results/sim_value.csv`).



Four results shape the concept. A ballpark is enough: a gauge with 8.6% median error is worth €246, as much as the dataset's 1.3% gauge at €239, and value only drops at 17% error, to €186. Reading low hurts while reading high does not, which is why the stroke count is set 10% high, and that setting lifts the plain stroke count from €198 to €237, 99% of the gauge. Lost telemetry is the one stroke failure that hurts badly: 10% of days lost unnoticed gives €112, while catching it and falling back to fill gives €174. Finally, pressing habits decide everything. When sites press 1 to 20 times differently, one average per stream gives only €78, while teaching each site with 3 weighed lifts gives €225 with an 8% site error and €218 with 12%.

The fill reading cannot replace the strokes. Kilos estimated from the fill reading give €147, and calibrating every fill sensor at the factory does not help, at €140, because the fill reading is too coarse to carry mass.

SMART workflow



Every record leaves SMART as a FIWARE Smart Data Models `WasteContainer` entity in NGSI-LD (`analysis/fiware_export.py`). The weight carries its method (`estimated:stroke-count`), its confidence, the kilos per stroke and the stroke count, so a city, a recycler or the customer

can always see that the figure is an estimate and how it was made. Records for all 447 machines pass the schema, apart from the location, which the synthetic sites lack. Of the 447, 23 are marked low confidence for telemetry gaps and 3 as over-reads.

The rules on a year of SMART data

`smart/replay.py` runs the rules above hour by hour on the dataset's telemetry for all 447 machines and 17,842 fill cycles. Each hourly value uses only the readings up to that hour, so the same code can run as a live service in the SMART cloud. The replay uses one kilos per stroke per stream, because in this dataset every site presses alike, and there per-stream and per-site values agree (6.5% and 7.5% median error in `results/habits_eval.csv`).

The replay shows when lifts really happen, measured in kilos (`results/replay_by_trigger.csv`). On-call lifts ordered by the fill reading come at a median of 80.4% of capacity, and 49.2% of them before the 80% call. Fixed-schedule lifts come at 51.9%, and 83.0% of them before the call, so the 4,752 scheduled fill cycles lift containers that are about half full. Maximum-interval lifts come at 54.5%, which may be contractual or hygienic and cannot all be avoided.

Against the official labels, the strokes hold up well. Of the 1,854 labeled unnecessary lifts, 1,777 end a complete fill cycle, and at 1,713 of those (96.4%) the stroke estimate was below the 80% call, so SMART Weight would have held them. They carry €253,091 of the €262,849 at stake. Holding a lift is not the whole saving, because the next lift still comes, which is why the simulator, not this count, gives the net value. The rule is also selective: on on-call lifts it holds 97.2% of the labeled unnecessary ones and 43.0% of the others.

Customers who had to call for an extra lift show the other side. In 508 ad hoc customer calls the median fill reading was 96.9%, and in 79.3% of them the kilo call had already fired, a median of 49 hours before the lift. The watchdog fired in 274 of the 312 labeled stuck-sensor periods (87.8%), and 85.4% of its alarm hours fell inside such a period.

What the data makes possible for Europress

Once SMART Weight runs, Europress collects more than a weight per lift. It learns how hard and how often each machine is used, how each site presses, when strokes and fill disagree, and how all of this moves with the seasons. This data serves Europress's own service, sales and product teams as well as its customers and partners. Some of the uses below are tested in this work, and the others are marked as ideas.

Service planned by use, not by calendar. Machines differ hugely in how hard they work. Over the year, the busiest tenth of machines pressed 10,260 times or more, while the quietest tenth pressed 830 times or fewer, a 12.4-fold spread (`results/data_uses.json`). With such a spread, a fixed service interval comes too early for most machines and too late for the busiest ones. Planning ram seals, hydraulic oil and inspections by stroke count would put service visits where the wear is, which lowers Europress's service cost and avoids breakdowns that customers notice. The synthetic data does not link breakdowns to use (corrective work orders range from 1.18 to 1.41 per machine across the four use quartiles), so this benefit is untested here. The stuck-sensor watchdog is the tested part of

the same idea: it turns a disagreement between strokes and fill into a service ticket before the customer pays for empty lifts.

Sales leads from the machine's own behavior. The kilos at each lift show which machines do not fit their sites. In the replay, 88 of 447 machines are usually lifted below half their capacity by the stroke estimate, while 46 are usually past full. The first group may be over-serviced or too large, the second too small. Both are concrete reasons for a sales call: a different container size, a second machine, or a move from a fixed schedule to on-call lifting by kilos, which the 4,752 half-full scheduled lifts above make a clear case for.

Better service to the customer. The customer gets a kilo record per lift and a monthly report of kilos and cost per tonne by waste stream, which no operator invoice gives today. Pressing habits are part of that report. A site that presses after every box wears the machine and uses energy for little gain, and a short note to the site manager can change that. The drift alarm also shows when a site's habit has changed, for example after a new tenant or new staff.

Weather and seasons. Rain adds water mass without strokes, and holidays and summer shutdowns stop the strokes altogether. The dataset's site calendar records closures, and the Finnish Meteorological Institute publishes its data sets freely for public use, so both can feed a kilo forecast per site. In this dataset, sites that store material outdoors show kilos per stroke that differ from indoor sites by 3.4% or less in every stream (`results/data_uses.json`), so the rain effect is too small to see here and the forecast is untested.

Product development. Real stroke counts, pressing habits and fill curves per waste stream tell Europress's engineers how machines are actually used: how often the ram runs half empty, which hopper sizes fit which sites, and where a pressure-based detector for empty presses would pay off. This feedback costs nothing extra once the data flows.

Partners and new data products. Haulers could plan routes and truck loads from kilos per site ahead of each lift, recyclers could see incoming material by stream, and cities get the same records through the open FIWARE format. Aggregated and anonymized kilo flows by region and stream could become a data product of their own. These uses depend on customer consent and on what customers allow Europress to share, which is a contractual question, not a technical one. All of them are untested.

What stays out of reach. Disputing an operator's invoice needs about 1.5% accuracy, which the stroke count does not reach. Customers who need it can buy a reference gauge, which turns a limit of the concept into an option to sell.

Product, revenue and rollout

SMART Weight is sold as a software add-on for connected machines. It bundles lift timing by kilos, the stuck-sensor watchdog and a kilo record per lift with method and confidence, and each site is taught once at setup. The price is a hypothesis: at €225 per machine-year of simulated value, a third of the saving would be about €75 per machine-year, and the simulated customer value per 1,000 connected machines is about €225,000 a year at the dataset's prices.

Europress earns in four ways. The subscription is recurring revenue on machines already sold. The watchdog and use-based service plans bring service work to Europress's own technicians. The right-sizing leads and the move from schedules to on-call lifting create sales. Reference gauges become an option for customers who need tonnage checks. In return Europress provides the calibration kits or ticket requests, a few reference gauges, and the upkeep of the models.

The rollout keeps the risk low at each step.

1. **Desk check, one day.** Run `analysis/strokes_eval.py` on Europress's real cycle counts and shared tickets. Pass mark: median error at or below 9% per stream.
2. **Pilot, 8 to 12 weeks.** 20 to 30 machines in plastic film, mixed plastic and mixed waste, where the value is highest. Pass marks: the targets in `docs/concept.md`, fewer lifts per tonne and no added overflow.
3. **Launch on connected machines.** A software update, with each site taught by 3 tickets or the kit. Plastics and mixed waste first, then cardboard, paper and biowaste.
4. **Reference network.** A few gauged machines per stream and region keep the stream averages up to date.
5. **Unconnected and other-brand machines.** A retrofit unit with a motor-current stroke counter, after its own pilot.

The main risks and their answers are known. Real controllers might not report strokes, which the desk check settles on day one and a firmware release fixes. Operators paid per lift lose lifts, so the sales story is fuller trucks and a lower cost per tonne, starting with customers who order their own lifts. A customer might treat the estimate as a billing weight, so every record says "estimated", and billing stays on the weighbridge. Real sites may press more erratically than any scenario tested here, which per-site teaching, the drift alarm and the pilot address. Sites waiting for their 3 weighings run on the fill reading or the gradient-boosting fallback until they are taught.

Where AI is used

In the product, kilos per stroke are learned per site from 3 weighed lifts, starting from a stream average learned on reference machines. A drift alarm and a stuck-sensor watchdog compare mass with volume, and every figure carries a confidence. Gradient boosting that fuses strokes, the fill reading and a Kalman estimate is kept as the fallback for sites not yet taught, at about 15% median error.

Three methods were tested and not adopted: a Kalman filter that learns each site from the fill reading alone, inverse-variance fusion, and conformal 90% ranges, which covered only 86% to 87% of true weights. The next step is reading each stroke's hydraulic pressure curve. The point where pressure starts to rise shows how big the charge was, and a press on an empty hopper never builds pressure, so the curve measures charge volume and flags empty presses. A model can learn this from lift totals alone, because the strokes of one lift must add up to its weight (learning from aggregate observations, arXiv 2004.06316). The dataset contains no pressure curves, so this remains untested.

What this work rests on, and what it does not show

What it assumes

The concept assumes that real SMART controllers log the number of compaction strokes, as the dataset's `compaction_cycles` field does. Every stroke is started by the controller, so a machine generation that does not report the count needs a firmware change, not new hardware, but this has not been confirmed with Europress.

It also assumes that each site can be taught once with 3 weighed lifts, either from 3 weighbridge tickets asked for at setup or from a mobile calibration kit, and that a site's pressing habit stays stable between drift alarms. The waste stream configured in SMART is assumed to be correct, because it sets the start value before the site is taught.

The kilo trigger, a lift at 80% of the container's capacity in kilos, is the rule Europress's simulator uses. Capacity in kilos comes from the generator's own formula, volume x compacted density x (1 + compressibility / 2).

Where the evidence is weaker than it looks

All data is synthetic, and all euro figures use the dataset's hypothetical prices. The simulator creates strokes from waste deposits with a fixed mean mass per deposit for each stream, so stroke counting works there partly by construction. The pressing-habit stress test exists to break that, but its habits are injected by this work, not observed, so only a pilot can show how far real sites differ. The scenario where sites press less often per kilo thins strokes at random, which overstates the error of sites that wait for a full hopper.

In the simulator, site habits stay stable and the error left after teaching a site is a fixed spread of 8% or 12%, taken from the stress test. The value of the drift alarm under changing habits is not in the euro figures. The stroke estimate in the simulator also starts each fill from the true leftover mass after the previous lift, which is a small advantage over real operation.

Several settings were chosen by hand on this dataset: the 1.1 safety factor, the stuck-sensor watchdog (fill at 85%, kilos under 40%, 6 hours after a lift, picked from three tried settings), the drift alarm (2 times for 3 lifts, picked from four tried settings), the Kalman drift setting and the gradient-boosting settings. They need confirming on real data. The watchdog's alarms fall outside labeled stuck-sensor periods in 14.6% of alarm hours.

The replay learns kilos per stroke on all machines, so it is in-sample, while the accuracy figures always learn on other machines. A partial lift does not reset the stroke count in the replay, so the estimate reads high until the next completed lift.

What the stroke count cannot do

The stroke count misses the brief's simulation target of about ±100 kg on a 3 t load: only 34.0% of loads between 2.5 and 3.5 t are within it. It is good enough to decide when to lift

and to report cost per tonne, but not to check an operator's tonnage, which needs a reference gauge or a weighbridge ticket. On all lifts, 1 in 10 is off by more than 61%, mostly overestimates, and the capacity check brings that to 24.7%. Underestimation by more than 20% happens in 2.5% of lifts, mostly where telemetry has gaps.

The conformal 90% ranges covered only 86% to 87% of true weights, so they are not used.

What was not priced

The mobile calibration kit, the reference gauges, their installation, kit transport, the firmware change and cloud compute were not priced. The kit price comes from one listed container scale. The technician time, kit life and kit use per year are assumptions, and the software effort is a guess.

What remains untested

- Stroke counting, per-site teaching and the drift alarm on real Europress machines.
- The mobile calibration kit, and gauge designs for COMBI feet and DUO rails.
- Reading each stroke's hydraulic pressure curve to size the charge and catch empty presses. It measures charge volume, not mass, and the dataset has no pressure curves.
- Heavy single items and rain, which make the estimate read low. In this dataset, sites that store material outdoors show no visible effect on kilos per stroke.
- Stroke counting from pump motor current on machines without a cycle counter.
- Use-based service planning, which the synthetic data cannot test because it does not link breakdowns to use.
- Weather and holiday forecasts of kilos, data products for haulers, recyclers and cities, and the customer consent they need.
- A live FIWARE context broker. The records were validated against the Smart Data Models schema only. The synthetic sites have no coordinates, so the records leave out `location`, which a site address or GPS fix would fill in production.